

# Lecture 00: Welcome and Introductions

COMS 4170 • Spring 2026

Prof. Celeste Layne  
Tuesday January 20, 2026



COMS 4170 · SPRING 2026

# Welcome & Introductions





## Celeste Layne

Adjunct Associate Professor,  
Columbia University

Urban planner by training — software engineer by profession. Adjunct faculty at Columbia for five years, teaching: *Methods as Practice/Practice as Methods* and *Computational Design Workflows* in the CDP Program and a Visual Studies course, *Coding for Spatial Practices*.



# COMS 4170 Team

## Prof. Celeste Layne

- Office Hours: Mondays 7PM to 8PM
- Please come to my office hours. Ask me anything:
  - Programming questions
  - Debugging help
  - Anything will help you learn in this class I will attempt to address in a helpful and judgment free way

## 9 Teaching Assistants (TAs) for approximately 200 students

- Each of you will be placed into sections of approximately 20 students
- Get to know your TA and the other students in your section



# COMS 4170 Teaching Assistants



**Haiyue Zhang**

MS, Computer Science  
hz2995@columbia.edu



**Luana Liao**

BS, Computer Science  
luana.liao@columbia.edu



**Jenny Ma**

PhD, Computer Science  
jm5676@columbia.edu



# COMS 4170 Teaching Assistants



**Brandon Pae**

BS, Computer Science  
btp2109@columbia.edu



**Daniel Manjarrez**

BS, Computer Science  
dam2274@columbia.edu



**Kevin Gutierrez  
Garcia**

BS, Computer Science  
kmg2226@columbia.edu





# COMS 4170 Teaching Assistants



**Karthik Sreedhar**

PhD, Computer Science  
ks4190@barnard.edu



**Riya Sahni**

PhD, Computer Science  
riya.sahini@cs.columbia.edu



**Anushka Bhatnagar**

MS, Computer Science  
ab5920@columbia.edu



# Asking for HALP!

- Each of the TAs will have weekly office hours (TBD). Class will normally run **Tuesday, 7:00PM to 9:30PM**. My office hours will be on **Monday, 7PM to 8PM**.
- We will not respond to personal emails with code questions, but ask that you post them instead on [Piazza](#). We will answer questions there so that everyone can learn, ***we also strongly encourage you to answer each other's questions.***
- Some students are good at one thing, others at another. Leverage each other's strengths.
- Please feel free to write us with logical, conceptual or other questions.
- See the [HALP! Page](#) on Courseworks for additional resources.





# Asking for HALP!

- This term we will be using Piazza for class discussion. The system is highly catered to getting you help fast and efficiently from classmates, the TA, and myself.
- Rather than emailing questions to the teaching staff, I strongly encourage you to post your questions on Piazza.

**Find our class signup link at:**

**<https://piazza.com/columbia/spring2026/comsw4170/home>**



COMS 4170 · SPRING 2026

# House Rules



# Classroom Culture

- This is an open, safe environment.
- Keep criticism positive and constructive.
- Take ownership of your experience.
- Check your ego at the door.
- Channel empathy.
- Don't compare yourself to others, compare yourself to who you were yesterday.
- You get back what you put in.



COMS 4170 · SPRING 2026

# Class Business



# COMS 4170 Goals

1. We want you to be able to **build interfaces** that suit and needs and abilities of users.
2. We want to convince that **design is process for solving real world problems** in computer science and beyond.
3. We want to you be **engaged** in the class by interacting with the staff and your fellow students...not bored, lost, zoned-out or playing *Minecraft* during class.
4. We want you to make a habit of **thinking about users**.



COMS 4170 · SPRING 2021

# Why are user interfaces important?

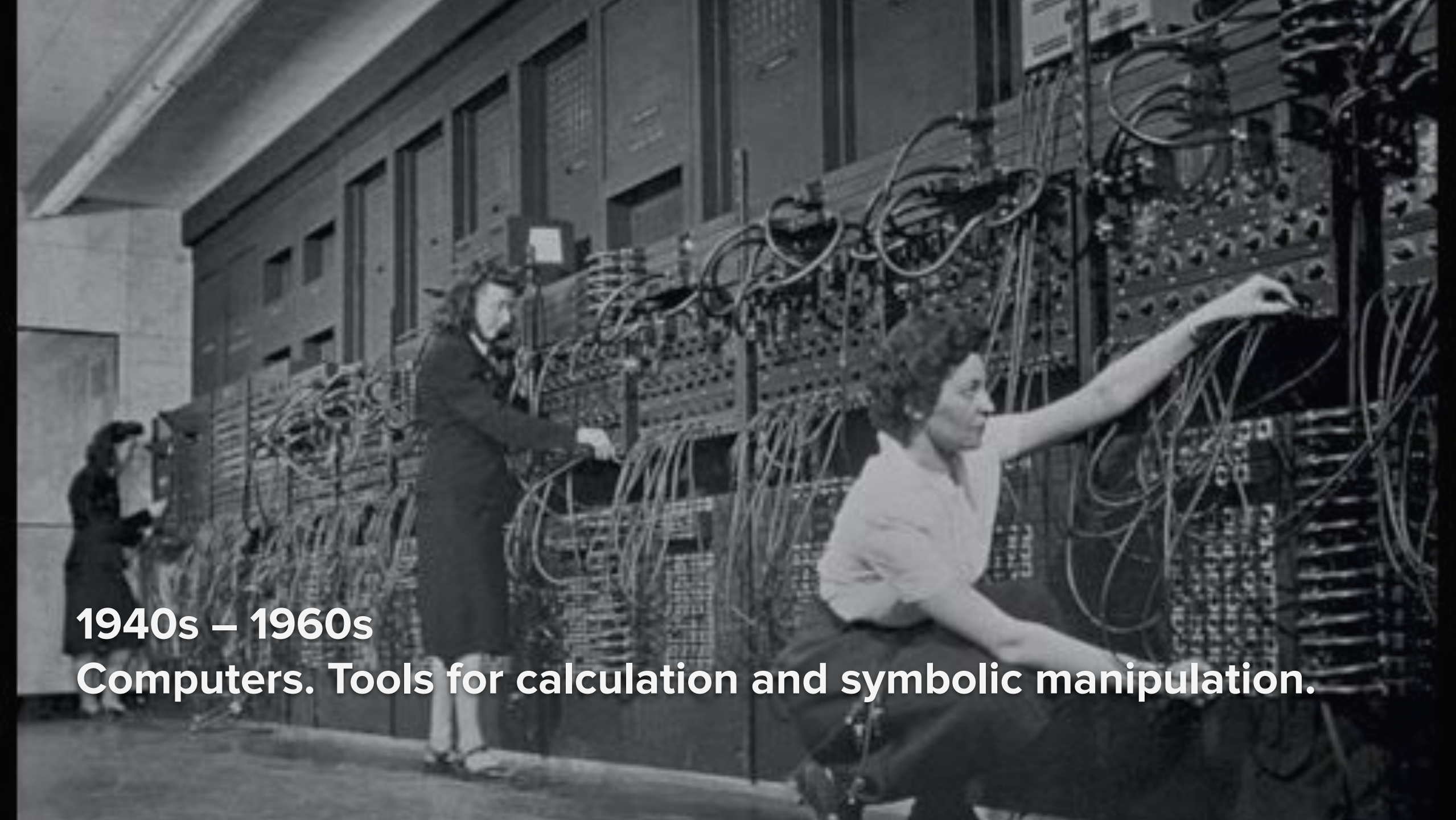






1613 – 1940s

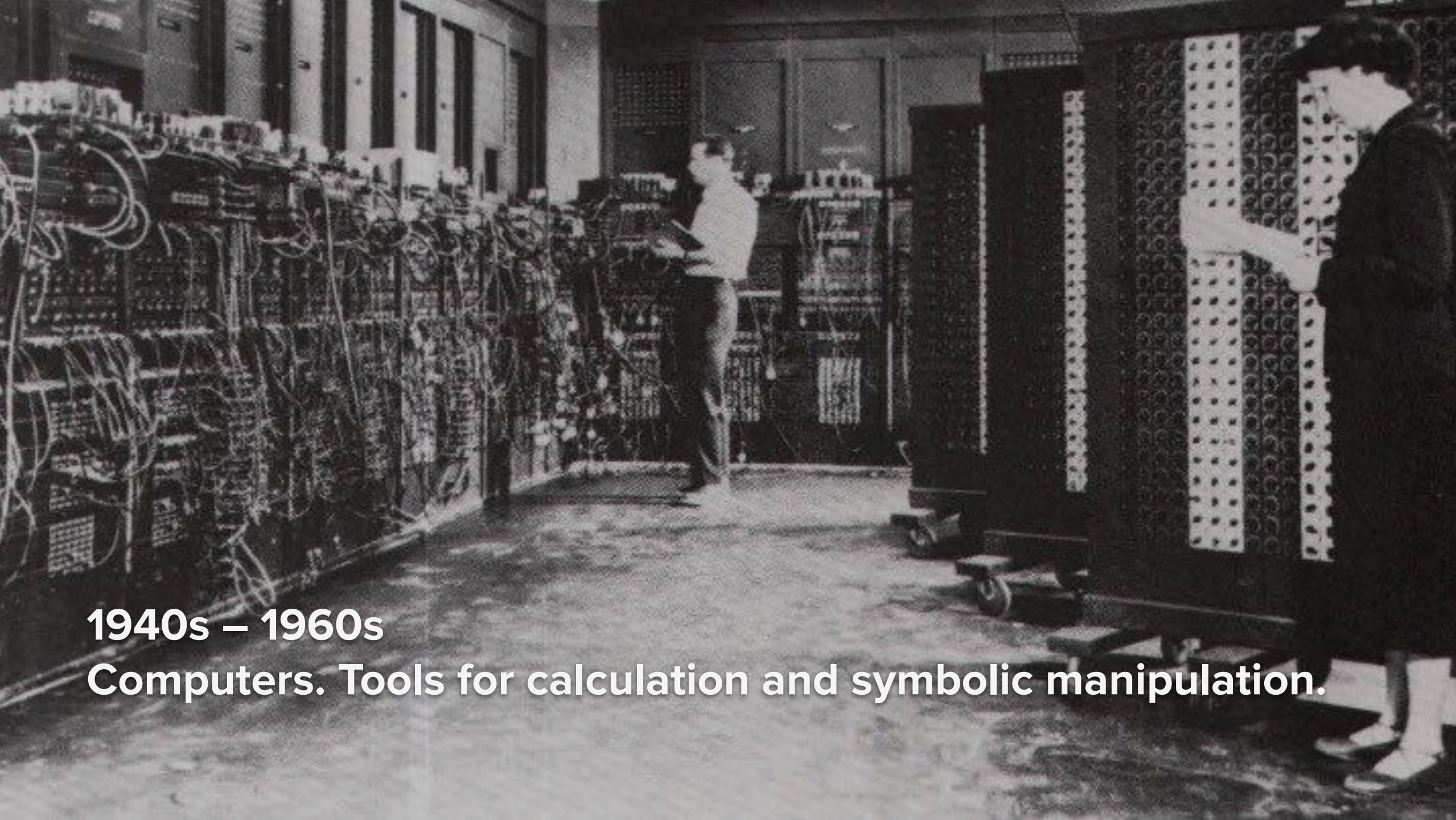
Computers. People who performed calculations.

A black and white photograph showing three women in a room filled with large, floor-to-ceiling computer cabinets. The cabinets are densely packed with electronic components and a complex web of cables. One woman in the foreground is crouching, reaching up to adjust a cable. Another woman stands in the middle ground, also working on the cables. A third woman is visible in the background on the left, partially obscured. The scene illustrates the scale and complexity of early computer systems.

**1940s – 1960s**

**Computers. Tools for calculation and symbolic manipulation.**





**1940s – 1960s**

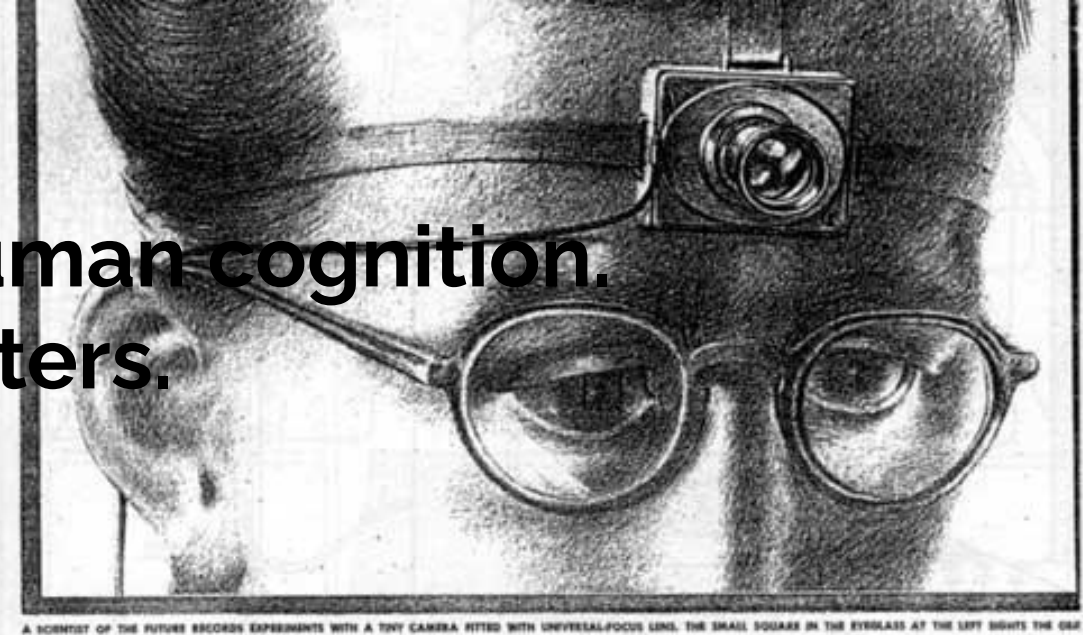
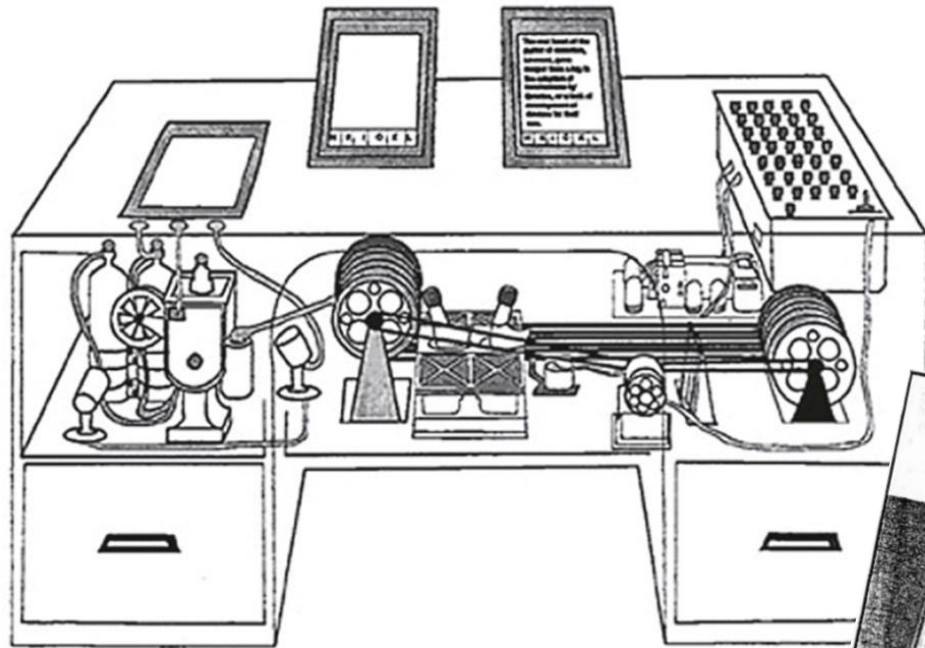
**Computers. Tools for calculation and symbolic manipulation.**



1945

# Computers. Tools to augment human cognition.

## Vannevar Bush's vision of computers.



A SCIENTIST OF THE FUTURE RECORDS EXPERIMENTS WITH A TINY CAMERA FITTED WITH UNIVERSAL-FOCUS LENS. THE SMALL SQUARE IN THE EYEGLOSS AT THE LEFT BRIGHTS THE GEAR

## AS WE MAY THINK

A TOP U.S. SCIENTIST FORESEES A POSSIBLE FUTURE WORLD  
IN WHICH MAN-MADE MACHINES WILL START TO THINK

by VANNEVAR BUSH

DIRECTOR OF THE OFFICE OF SCIENTIFIC RESEARCH AND DEVELOPMENT  
Condensed from the Atlantic Monthly, July 1945

This has not been a scientists' war; it has been a war in which all have had a part. The scientists, burying their old professional competition in the demand of a common cause, have shared greatly and learned much. It has been a war in which they have been able to work in effective partnership. What are the scientists to do

particularly for the medical scientists, there can be no work has hardly required them to leave the old on able to carry on their war research in their ries. Their objectives remain much the same. have been thrown most violently off stride, who s for the making of strange destructive gadgets, ew methods for their unanticipated assignments. on the devices that made it possible to turn back rked in combined effort with the physicists of our thin themselves the stir of achievements. They have am. Now one asks where they will find objectives

mountain of research. But there is increased evidence tged down today as specialization extends. The investi-the findings and conclusions of thousands of other work-ich he cannot find time to grasp, much less to remember, specialization becomes increasingly necessary for prog-

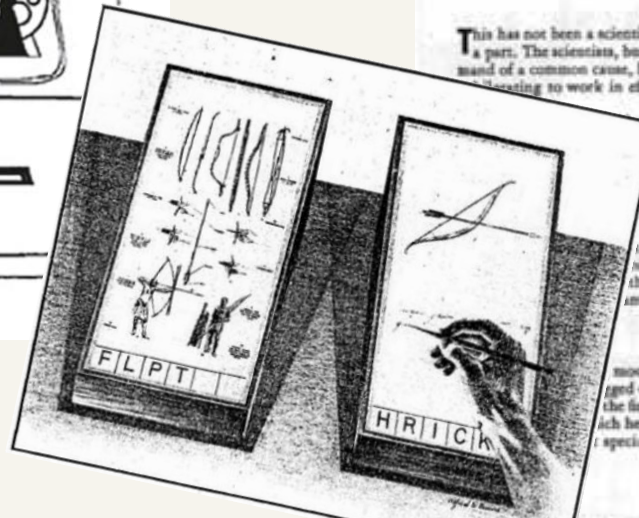
ress, and the effort to bridge between disciplines is correspondingly ssp ficial.

Professionally our methods of transmitting and reviewing the results research are generations old and by now are totally inadequate for their p pose. If the aggregate time spent in writing scholarly works and in readi them could be evaluated, the ratio between these amounts of time mlt well be startling. Those who conscientiously attempt to keep abreast of c rent thoughts, even in restricted fields, by close and continuous reading mlt well shy away from an examination calculated to show how much of the p vious month's efforts could be produced on call.

Mendel's concept of the laws of genetics was lost to the world for a g eration because his publication did not reach the few who were caps of grasping and extending it. This sort of catastrophe is undoubtedly be repeated all about us as truly significant attainments become lost in the n of the inconsequential.

Publication has been extended far beyond our present ability to make t use of the record. The summation of human experience is being expande a prodigious rate, and the means we use for threading through h the co quent mass to the momentarily important item is the same as was used the days of square-rigged ships.

But there are signs of a change as new and powerful instrumental come into use. Photocells capable of seeing things in a physical sens, vanced photography which can record what is seen or even what is t thermionic tubes capable of controlling potent forces under the guidanc



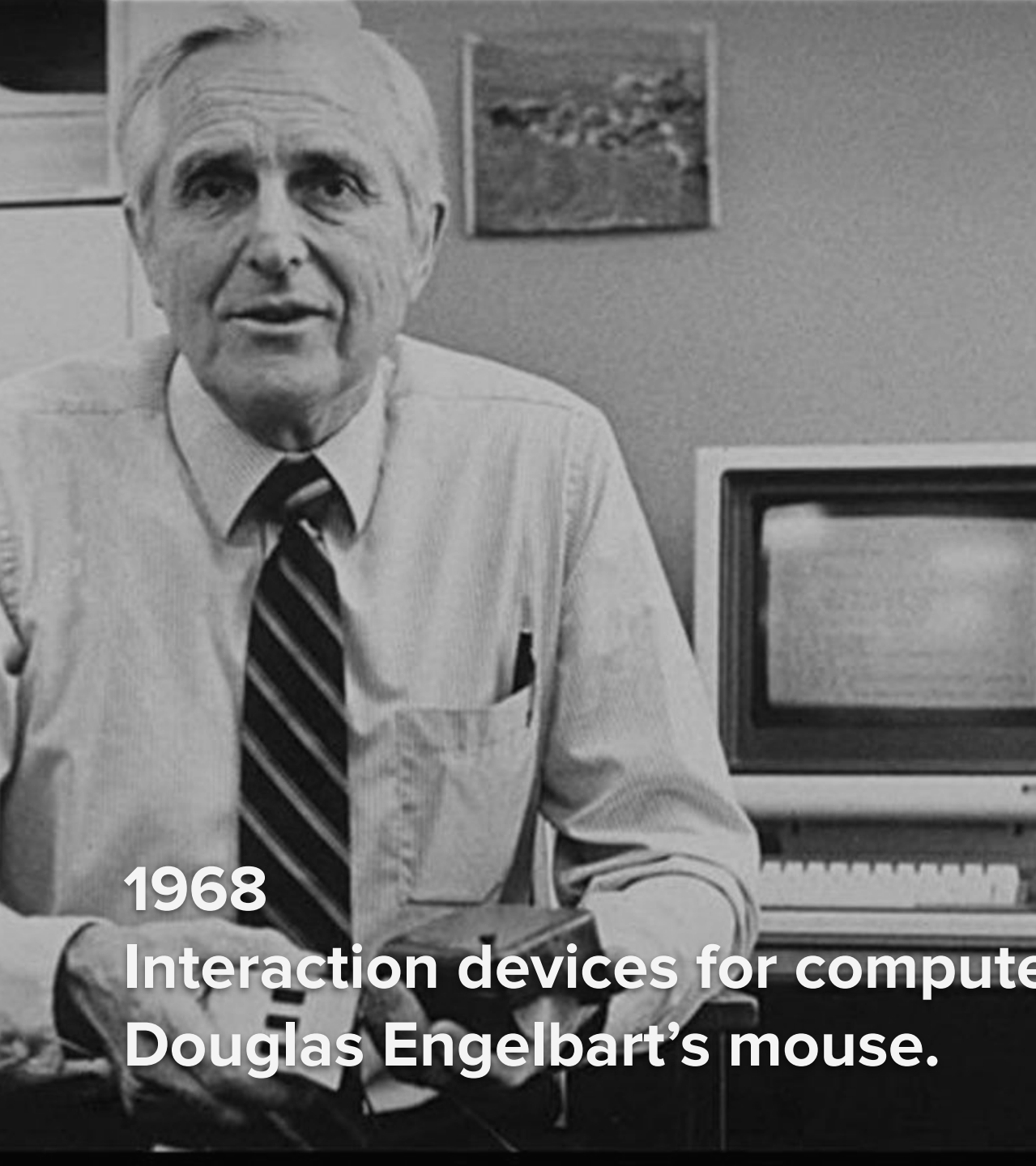


**1963**

**First Graphical User Interface.**

**Ivan Sutherland's CAD software, Sketchpad**





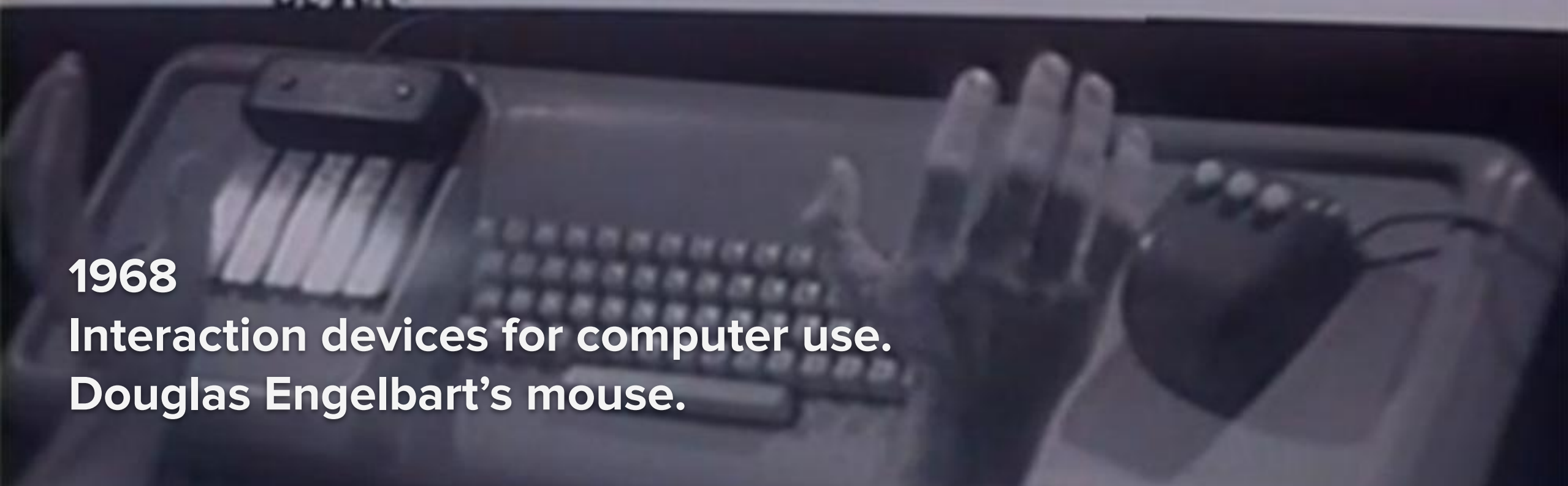
**1968**  
**Interaction devices for computer use.**  
**Douglas Engelbart's mouse.**



PRODUCE  
ORANGES  
APPLES  
BANANAS  
CARROTS  
LETTUCE  
PEARS

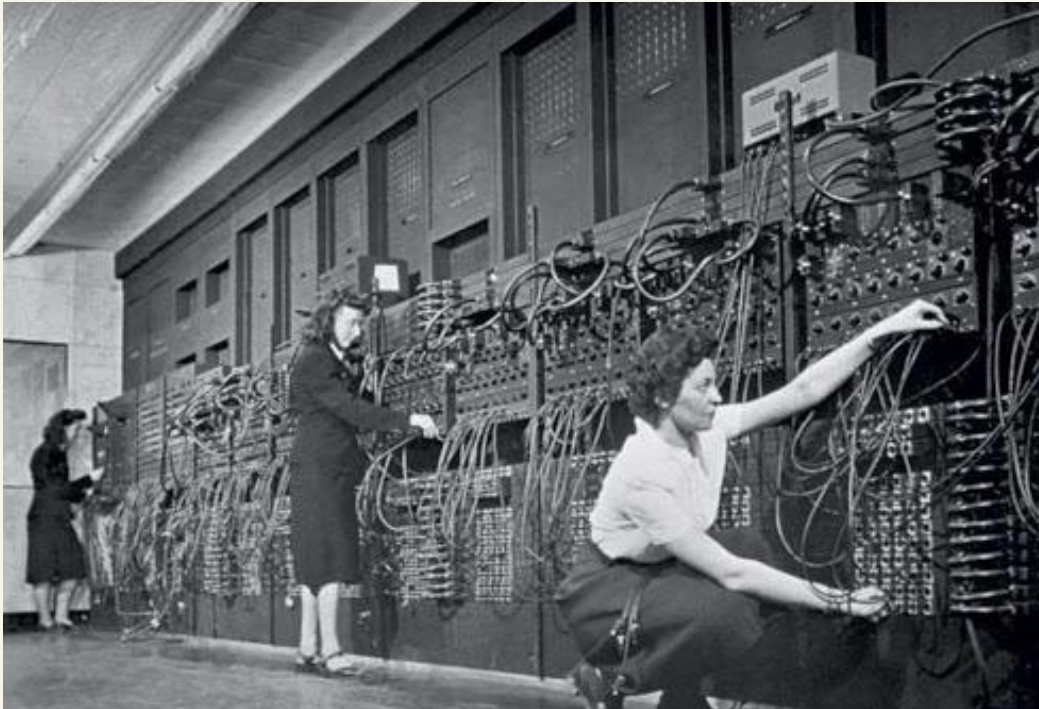
1968

Interaction devices for computer use.  
Douglas Engelbart's mouse.



# Computers.

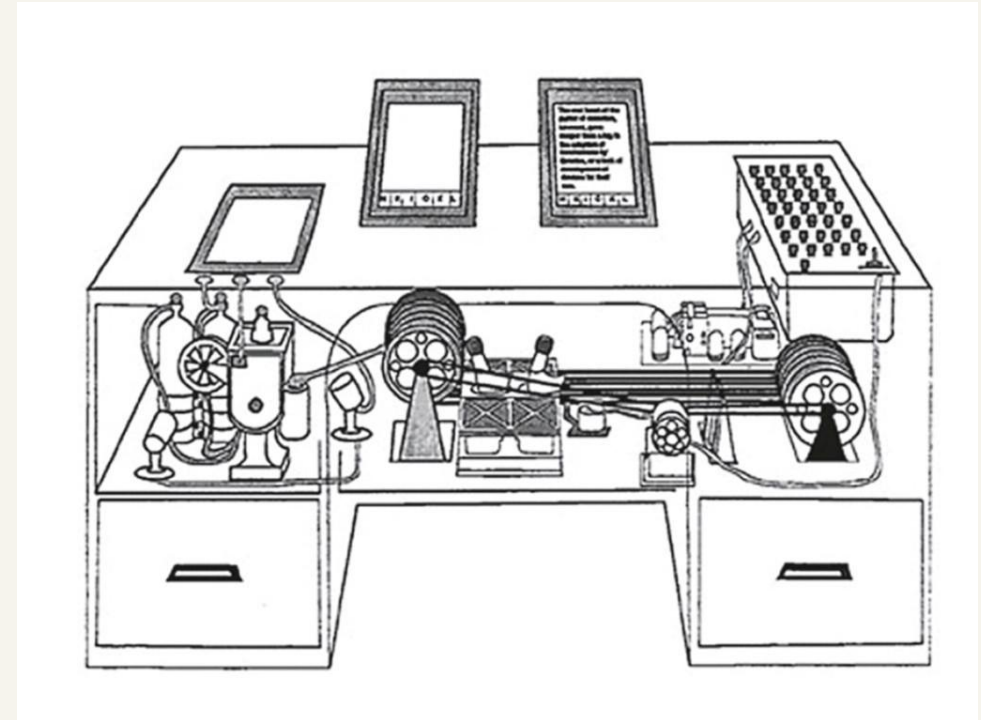
## Tools for calculation



Computer-centric interface

# Computers.

## Tools to augment human intelligence



Human-centric interface



# The Internet.

## The rise of usability.



For physical products, users did not get to experience the usability of the product until after they bought it.



For desktop software, users call expensive support centers, but the costs aren't “charged” to the software engineers, so they have no motivation to ship great UIs.



On the Web, users experience the usability of a site before they have committed to using it and before they buy it. UI is now the primary “selling point” of software.

COMS 4170 · SPRING 2026

# What will we learn?



# Part 1: Understand users and build websites that suit their needs and abilities

- Learn UI principles and apply them to web programming.
- Technologies include: HTML, JavaScript, CSS, JQuery, Bootstrap, Flask.
  - We will teach all these technologies. No web programming experience required.
  - General experience with programming and debugging is required.
- Weekly assignments.
- An individual midterm project.



## Part 2: Design Usable Systems through iteration and feedback

Group project to design and build a website that teaches a concept through interaction and feedback.

- Weekly assignments.
- Completed by the group, turned in individually.
- Everyone in the group must have a working copy of the code running on their machine.
- Weekly assignments build up to the final deliverable.





# You will be assigned to a TA Group

- Your TA is your mentor. Get to know them 🧐
- Your group will have ~20 students.
  - Some of them will be your project partners.
  - Get to know them 🧐
- You can also come to any office hour you like.



## Grading. This class is not curved.

90% ≤ A ≤ 100%

80% ≤ B < 90%

70% ≤ C < 80%

60% ≤ D < 70%

0% ≤ F < 60%

**No A+ grades will be given.**

This class is designed so everyone can get 100% by showing up and doing the work.



# Grade breakdown.

- **Weekly Homework (60%)**
  - 12 homework assignments
  - Each homework is worth 5% of your grade
- **Individual Midterm Project (10%)**
  - Due 11:59 PM (Friday 13th March 2026)
- **Group Final Project (20%)**
  - Due 11:59 PM (Friday 15th May 2026)
  - No late assignments accepted. Not even one minute late.
- **No final exam**
- **Design Section Preparation and Participation (10%)**
  - In the design half of the class (after spring break). You will mainly learn by presenting your work and iterating on feedback from your TA.
  - It is essential to come prepared and actively participate.
  - If there is a week where you cannot make the regularly schedule time for your section meeting, email your TA in advance of the meeting and suggest 3 times in the next two days to schedule a make up.



# Weekly Homework Deadlines

- Generally, assignments are due Mondays at 11:59PM
- Assignments are meant to take 5-8 hours
  - For people with web programming experience they often take 2-3 hours
  - If you are spending significantly longer on assignments, come talk to me or a TA. We can help teach you programming and debugging strategies.
- There is also a short **warm-up assignment** due Fridays at 11:59PM





# Weekly Warm Ups

- Warm ups help you get started early.
- With 200+ people in this class, we can't help everyone on the assignment the night before it's due.
- We at least need to get installation issues out of the way.
- They are short. No more than one (1) hour; if it takes you more than 1 hour, turn in whatever you have at 1 hour and finish the rest on the Main Assignment.
- If there is something you are having trouble with, email your TA early.
- If you honestly think this will impede your learning rather than help, email me, and we'll work something out.
- **Don't use late passes for warm-ups.** If you are generally on time with warmups, that's what we care about.



# Late policy for homework.

- **Two (2) Late Passes:** You may turn in **two (2) main assignments up to 5 days late** with no penalty or excuse needed.
- Each late assignment must be turned in no more than 5 days (120 hours) after the original deadline. There is no difference between an assignment being 1 hour late or 99 hours late.
- If you are not using a late pass, late assignments will get 10% deduction for every 24 hours late.



# Cheating Policy

*Can I use generative AI?*

*Can I use code from the web?*

*Can I work on assignments with my friends?*

**Yes. Do it.**

**...as long as you can demonstrate understanding, ownership of the code and designs you produce in this class.**

If we have questions about your submission, we reserve the right to have you explain your code and thought process to us **in person**.



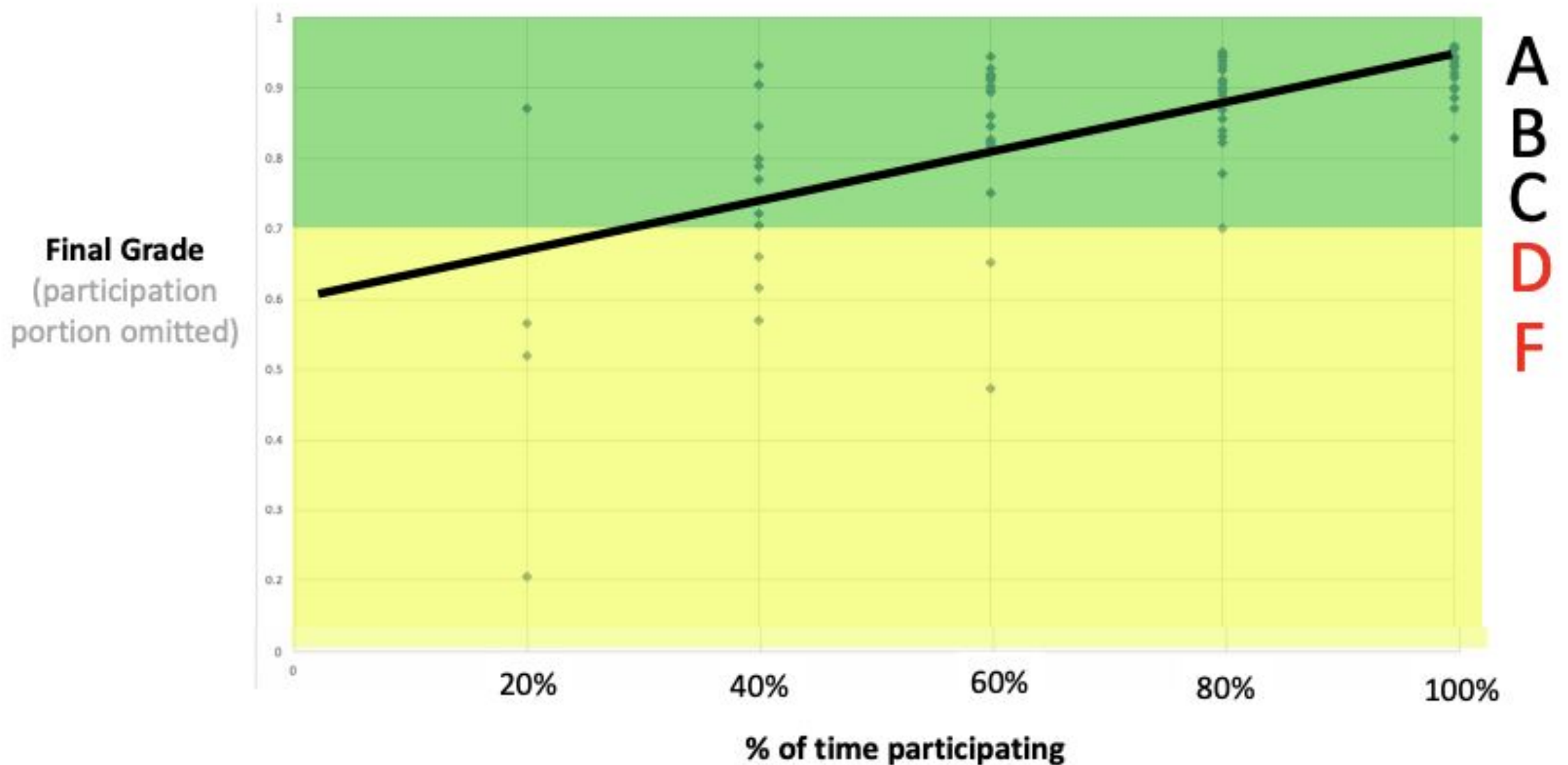
# Participation.

## How we measure participation (10% of the grade)

- During the second half of this class, you'll have weekly feedback sessions with your TA.
- These are only effective if you come prepared and participate.
- So there are points for it.
- If you can't make a feedback session, email your TA in advance and suggest 3 times in the next two days to schedule a make up.

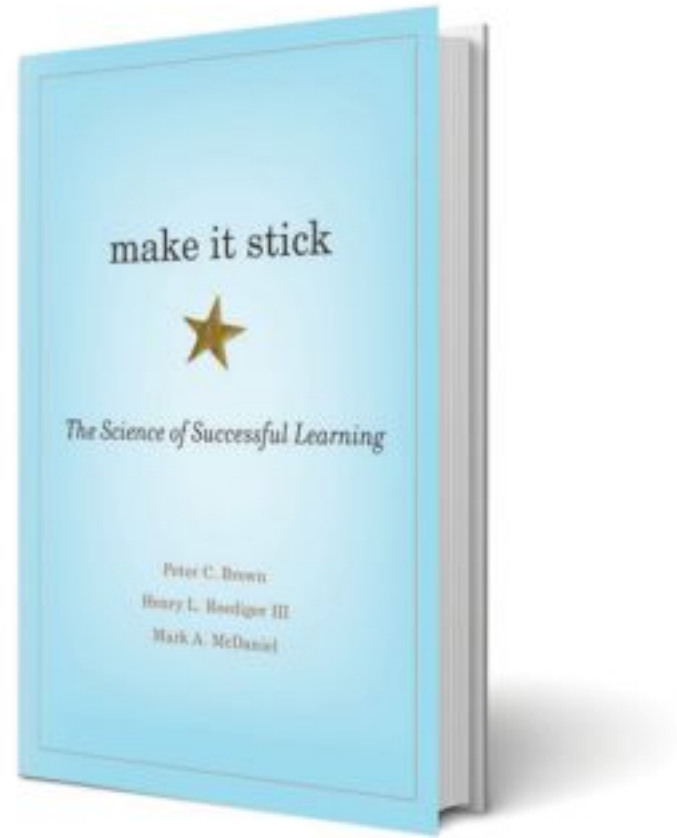


# Why is Participation 10% of the grade?

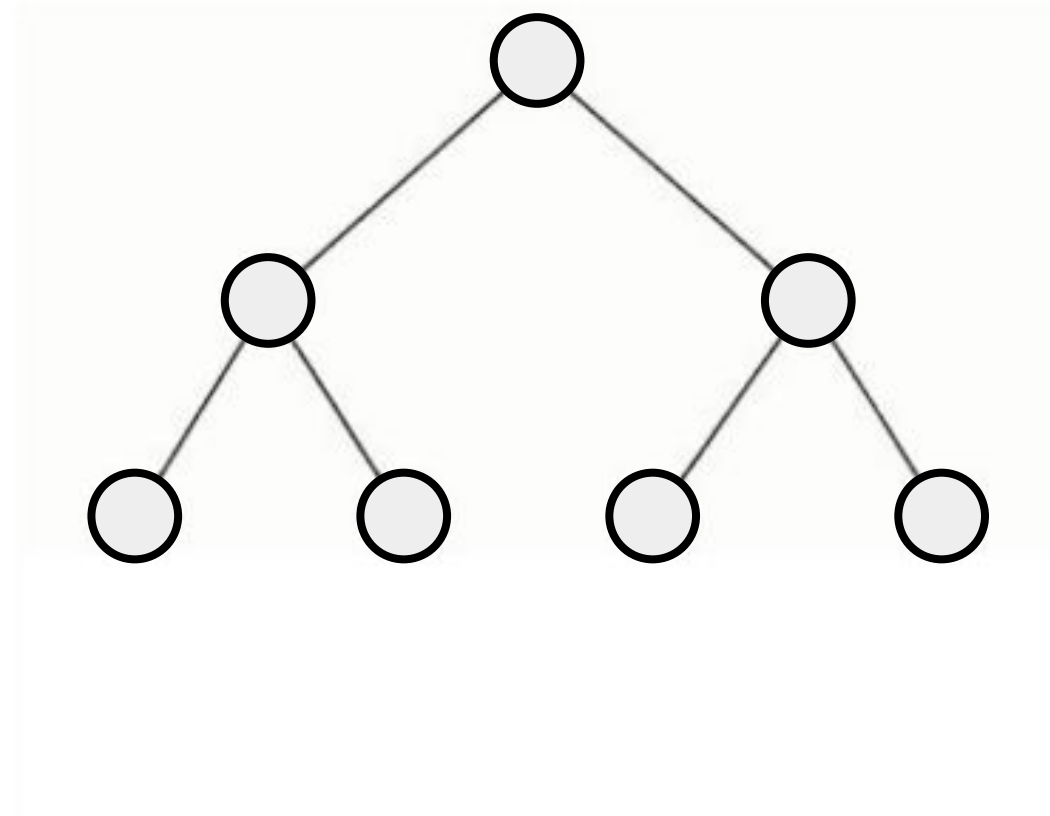




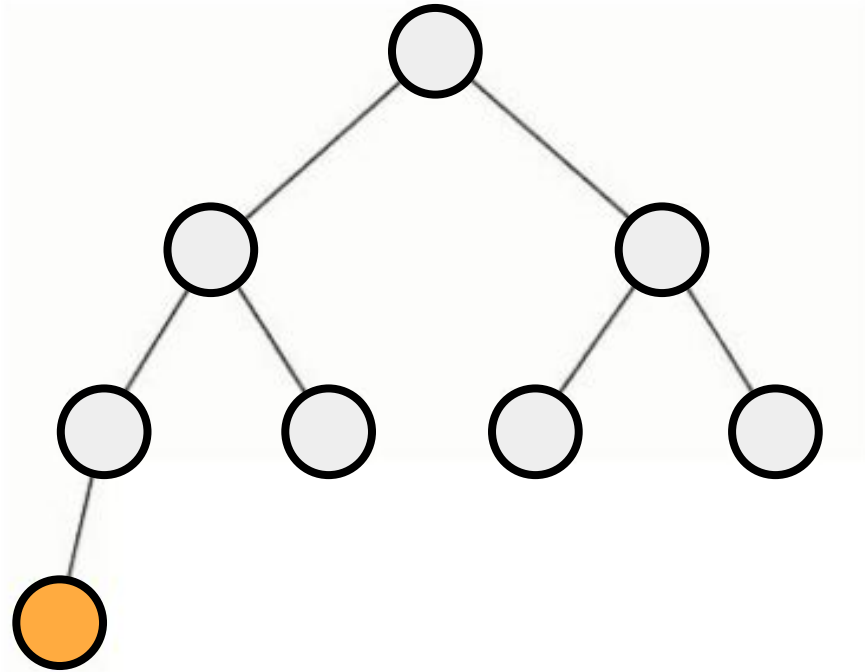
# Two reasons why participation affects learning



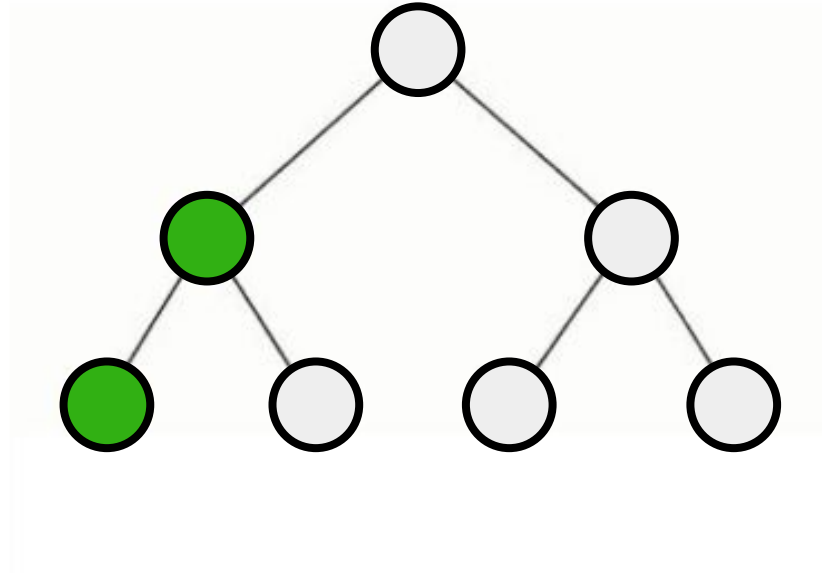
# Human memory is tree structured



**New knowledge gets appended to the tree**

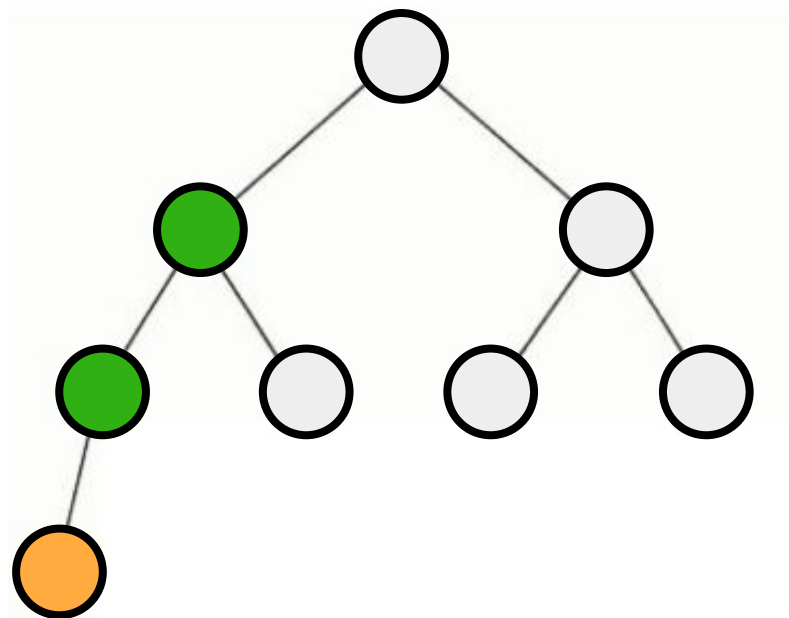


**Where on the tree does new knowledge gets appended?**



**...to where nodes are currently active.**

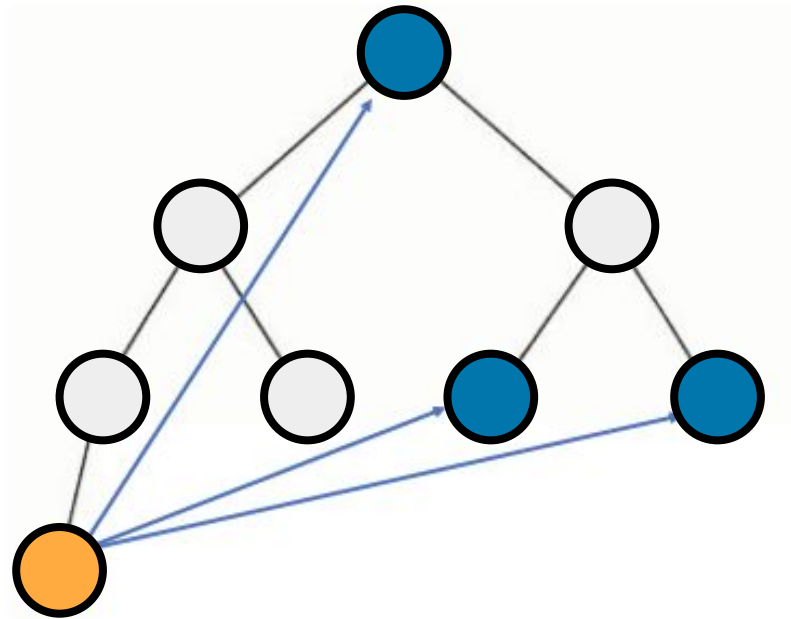
1. By guessing about new knowledge before it's presented, you **warm up** the right place to store it in memory.



**Generation:** Guessing before you hear the answer



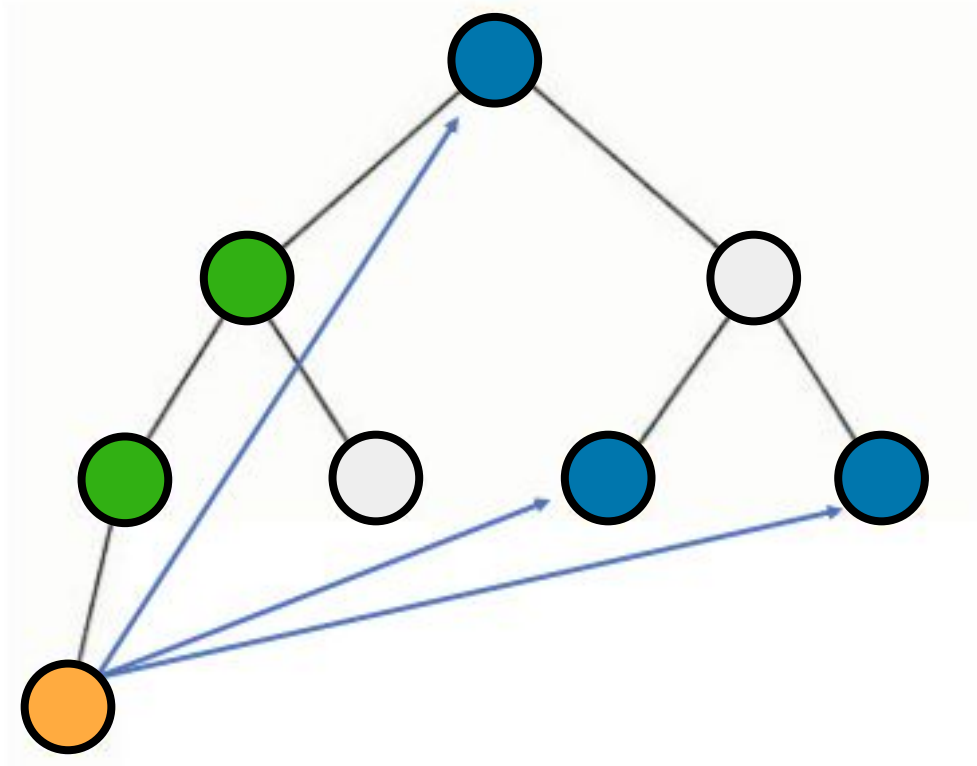
1. Once you hear new knowledge, connect it to other knowledge so it will trigger at relevant times.



**Elaboration:** Relating new knowledge to old topics

# Generation and Elaboration

Guess about the new knowledge. Must take risks, you will probably be (partially) wrong



Relate new knowledge to old topics. This aspect of participation is about providing insights.

# Our goal is to help you learn useful skills and a profound way of solving real problems.

- We want you to make a habit of **thinking about users**.
- We want to convince that **design is crucial** to Computer Science, to the universe and to everything
- We want you to be **engaged** in the class. Interact with us and your fellow students.



